

CHE 311 Homework Week 3

1. Velocity profiles and the stress components

For each of the following velocity distributions, find the total momentum flux $\Phi = \pi + \rho \mathbf{v}\mathbf{v} = \tau + p\delta + \rho \mathbf{v}\mathbf{v}$ (where τ is shear momentum flux, $p\delta$ is pressure, and $\rho \mathbf{v}\mathbf{v}$ is the convective momentum flux) and the divergence of Φ , which is $\nabla \cdot \Phi$. For all the cases, $p = -\rho gy$, where ρg are constants.

- (a) $v_x = by, v_y = 0, v_z = 0$
- (b) $v_x = by, v_y = bx, v_z = 0$
- (c) $v_x = -by, v_y = bx, v_z = 0$
- (d) $v_x = -0.5bx, v_y = -0.5by, v_z = bz$

2. Calculate the force

A 5% aqueous solution of polyethylene oxide (molecular weight 20,000) at 333 K has power law parameters $n = 0.6$ and $m = 0.5 \text{ Pa}\cdot\text{s}^n$. This solution was filled between two parallel plates with distance 5 mm. The top plate was moving at a constant speed $U = 1 \text{ cm/s}$. Please calculate the shear stress on the plates.